

In-depth summary: adversarial-systematics VBF DNN loss-variant study

Prepared 2026-08-11 from the reports and supporting artifacts under `.agent-system/tasks/adversarial-syst-vbf-dnn-loss-variants/`.

Bottom line

The study does not determine whether the proposed adversarial-loss Variants A and B improve VBF DNN significance. The only two completed variant runs used a defective configuration in which gradients flowed through the nominal prediction inside the consistency loss. That created a confidence-inflation or “score-collapse” channel, especially severe under the high-score cut.

The corrected stop-gradient experiment was started but stopped after one of four folds, before Stage 2, template production, or significance evaluation. Consequently:

- The measured degradations should not be interpreted as evidence against Variants A or B.
- The predecessor study’s negative conclusion is also called into question because it used the same undetached formulation.
- The most defensible output is the diagnosis of the loss defect, not a physics-performance conclusion.

This is stated explicitly in the task’s `final-summary.md`.

What the study intended to test

The task followed up on `adversarial-syst-robust-vbf-dnn`, whose original adversarial objective degraded Run-2 pre-fit significance monotonically with increasing λ .

The proposed variants were:

- Variant A — consistency only: retain only $\lambda \times \sum \text{BCE}(\text{pred_variation}, \text{pred_nominal})$, dropping the label-pulling term and the factor of two.
- Variant B — tail restriction: apply that consistency penalty only where $\text{arctanh}(p_{\text{nominal}}) > 2$, equivalent to $p_{\text{nominal}} > 0.964$.

The rationale was sensible: the discarded label term fights discrimination, while most significance comes from the high-score tail, so applying decorrelation pressure over the full score distribution may waste classifier capacity.

The study inherited rather than recomputed:

- The 17-bin score binning.
- Three rebinned baseline seeds.
- Run-2 reference significance: 1.37093.
- Seed-noise scale: 0.00881.
- Required +2% floor: 1.39835.
- The keep/drop-degenerate augmented datasets.

The intended acceptance standard was demanding: at least three lambda values, a complete train → Stage 2 → Stage 3 → combine chain, and ultimately a full 26-variation result showing pre-fit significance above 1.3984 and beyond the seed-noise band.

Lambda calibration

The lambda values were calibrated from measured penalty magnitudes on 20,000 real events, rather than copied from the predecessor:

Configuration	Raw penalty	Events retained	Lambda at nominal-loss parity
Original/default loss	3.5475	100%	0.0796
Variant A only	1.1947	100%	0.236
A+B, cut 1.5	0.6194	16.9%	0.456
A+B, cut 2.0, no smoothing	0.3385	9.4%	0.834
A+B, cut 3.0	0.1315	1.6%	2.147

For keep-degenerate inputs, parity was measured as 0.816 rather than 0.834, only a 2% difference. The initial keep grid was therefore 0.082, 0.245, and 0.82, corresponding approximately to 0.1×, 0.3×, and 1× parity.

After the catastrophic 0.082 result, the high point was dropped and 0.005 was added to probe the very weak-coupling region. The calibration and decision history are in assumptions.md.

Completed significance results

Only two A+B points completed the four-fold training and four-variation significance chain. Both used:

- Keep-degenerate inputs.
- cut = 2.0.
- Consistency smoothing disabled.
- detach_nominal = false.
- Four variations: Total_up/down and mu_roccor_up/down.

Run	Pre-fit	Stat-only	Systematic headroom	Versus reference
Inherited gate reference	1.37093	1.52348	11.13%	–
Null/switch-off	1.37746	1.52449	10.67%	+0.48%
A+B, lambda=0.005	1.34935	1.50283	11.37%	-1.57%
A+B, lambda=0.082	0.82771	1.34432	62.41%	-39.62%
A+B, lambda=0.082, stop-gradient	–	–	–	Not completed

The lambda=0.005 point is 0.0281 below the null, about 3.2 sigma_noise relative to that null. Relative to the formal reference, the goal checker reports -2.4 sigma_noise.

A nuance in the written conclusion: stat-only significance did decrease at both lambda values, which is one part of the desired “pre-fit up/stat-only down” signature. The failure is that pre-fit significance also decreased, and systematic headroom worsened. Thus neither point satisfies the full conjunctive signature; “every component moved the wrong way” is somewhat too broad.

Per-year behavior reinforces the failure:

- At lambda=0.005, 2018, 2017, and 2016postVFP pre-fit values declined relative to their references; 2016preVFP improved, but not enough to rescue Run 2.
- At lambda=0.082, every year degraded sharply.
- The most extreme headroom was 93.5% in 2018 at lambda=0.082.

The raw values are in significance_summary.json.

Why the loss collapsed

The central defect is mathematically credible and is also consistent with the observed templates.

The loss was effectively:

`BCE(logits_variation, p_nominal)`

but p_nominal remained attached to the computation graph. Therefore the optimizer could change both:

- The varied prediction, which was supposed to match the nominal target.
- The nominal “target” itself.

For a probability target p , BCE at agreement is not zero; it equals the binary entropy $H(p)$. That entropy is minimized as p approaches 0 or 1. Consequently, the loss can reduce itself by making predictions more extreme instead of making the varied and nominal predictions genuinely stable.

For high-score events, the gradient through the target pushes the nominal score upward. Variant B selects only $p > 0.964$, removing the corresponding low-score/downward population that might otherwise partially balance the effect.

The hard selection mask is detached within a forward pass, but it is recomputed from the evolving model each batch. This can create positive feedback: events pushed over the threshold are subsequently included in the penalized tail.

Disabling consistency-term label smoothing made this worse. Smoothing had imposed a finite target cap near 0.98; removing it exposed the unbounded confidence-inflation direction. Accordingly, the earlier decision arguing that smoothing should be disabled is explicitly retracted in assumptions.md D-8.

Template evidence for the diagnosis

The $\lambda=0.082$ run did not create more total background; it moved background into the significance-bearing tail:

2018 SR bins	Null background	$\lambda=0.005$	$\lambda=0.082$
-----+-----+-----+-----			
Bulk, bins 0–13	3337.05	3334.94	3311.31
Tail, bins 14–16	23.91	26.02	49.65
Top bin only	10.73	12.14	36.12

At $\lambda=0.082$:

- Top-bin background increased by about 3.4×.
- Top-bin signal increased by only 1.7×.
- Top-bin S/B fell from 0.265 to 0.136.
- Tail purity fell in every high-score bin.
- Quadrature $S/\sqrt{B}$ fell by 12.9%.
- The resulting concentration in a DYVBF-heavy bin made the JEC templates

especially damaging, yielding 62.4% systematic headroom.

This is strong evidence that the loss was sharpening and migrating scores rather than decorrelating them.

The unfinished corrected experiment

The proposed correction was `--adversarial-detach-nominal`, which turns `p_nominal` into a fixed target while retaining gradient flow through the varied branch and shared model weights.

A directly comparable run at $\lambda=0.082$ was launched. It was stopped after fold 0 completed and fold 1 began:

- Fold 0 ended at epoch 24.
- Runtime before shutdown was approximately 57 minutes.
- No final manifest was written.
- No Stage-2 scoring or Stage-3 templates were produced.

- No significance result exists.

Its partial fold-0 diagnostics were encouraging but not decisive:

Fold-0 run	Best weighted validation AUC	Final nominal loss	Penalty change
Null	0.947003	0.28553	-12.6%
lambda=0.082, undetached	0.946595	0.29481	-32.2%
lambda=0.082, stop-gradient	0.946972	0.28550	-8.2%

The stop-gradient run tracked the null much more closely. However, the reports correctly warn that AUC is nearly blind to high-tail damage: the predecessor saw tiny AUC movement alongside a roughly 24% top-bin S/B loss. One fold therefore cannot answer the physics question.

The partial tag trained_advVarAB_lam0082_keepDegSG_Aug11_2026 must not be reused. The reports estimate a fresh four-fold training plus significance chain at roughly 2 hours 35 minutes.

Keep/drop-degenerate input issue

The task initially planned drop-degenerate inputs, but this was superseded by an analyst decision to prioritize consistency with Stage-2 inference.

- Keep-degenerate: reproduces Stage-2's current behavior, where the varied QvG inputs become constant -1.
- Drop-degenerate: removes those columns during training, but then trains against a perturbation Stage 2 does not actually apply.

The report concludes that neither is physically correct. A JEC shift changes which jet is leading in about 20.4% of events, so Stage 1 should populate each varied QvG branch using the QvG of the jet that actually leads under that variation. This remains the open upstream issue REV-001.

The raw logs also show an earlier, separate failure: the first drop-degenerate attempts at lambda=0.083, 0.25, and 0.83 aborted because batches with no selected high-score events returned a tensor without a gradient function. The current source contains an attached-zero fix, but those attempts produced no scientific results.

Proposed replacement if stop-gradient BCE still fails

proposal-shift-penalty.md proposes replacing BCE with a Huber penalty on displacement along the actual binned axis:

```
s = arctanh(p)
penalty = weighted mean of gate(s_nom) × Huber(s_var - s_nom)
```

Key elements are:

- Detached nominal score.
- Smooth, detached gate centered at $s=2.0$.
- Gate width around 0.25.
- Huber transition around 0.25.
- Normalization by selected event weight and number of variations.
- Optional lambda warm-up.

This would make the penalty exactly zero at agreement, prevent it from lowering its loss through confidence inflation, align the metric with Stage 3's score axis, and reduce domination by extreme QvG outliers.

The proposal's own limitation is important: it still penalizes motion within a wide histogram bin even when that motion does not change the statistical model. The suggested long-term endpoint is therefore a differentiable, soft-binned template-difference loss accumulated with an EMA. None of this has been implemented or measured.

Acceptance and artifact status

Against the formal criteria in task.json:

- AC1 not met: only two complete lambda points, both A+B and keep-degenerate; no three-point drop sweep and no Variant-A-only attribution.
- AC2 met: lambda calibration was measured and recorded.
- AC3-AC4 substantially met for completed points: the reports include pre-fit, stat-only, headroom, gate comparisons, and explicit signature evaluation.
- AC5 not triggered: no point showed the required signature, so no full-26-variation final run was warranted.
- AC6 met: setting and smoothing decisions are documented, including the later retraction.
- AC7 partially/formally met: a final summary exists, but the requested study itself is incomplete.
- AC8 not met literally: keep-degenerate Phase 1 failed, but the required drop-degenerate fallback did not complete before the analyst shutdown.

There is also a bookkeeping inconsistency:

- final-summary.md says the task was closed.
- current-state.json still says implementing.
- task.json still says created.
- iterations/001/ contains no generator, run, or reviewer reports.

- The claimed passing tests are not accompanied by captured test output in this task directory.

- The entire `.agent-system/` directory is currently ignored by `.gitignore`, so these artifacts are not tracked despite the canonical policy saying task JSON artifacts should be version-controlled.

Scientifically, the correct next action is exactly what the reports recommend: rerun the $\lambda=0.082$ stop-gradient configuration under a fresh tag, complete all four folds and the significance chain, and make every further decision conditional on that result.

Source artifacts reviewed

- `.agent-system/tasks/adversarial-syst-vbf-dnn-loss-variants/task.json`
- `.agent-system/tasks/adversarial-syst-vbf-dnn-loss-variants/current-state.json`
- `.agent-system/tasks/adversarial-syst-vbf-dnn-loss-variants/assumptions.md`
- `.agent-system/tasks/adversarial-syst-vbf-dnn-loss-variants/final-summary.md`
- `.agent-system/tasks/adversarial-syst-vbf-dnn-loss-variants/proposal-shift-penalty.md`
- `.agent-system/tasks/adversarial-syst-vbf-dnn-loss-variants/significance_summary.json`
- All files under the task's `logs/` and `scripts/` directories.